

Code - 3

Problem Statement: Write a program for Pre-processing of a Text Document: stop word removal.

```
In [1]: # Import libraries
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
```

```
In [2]: # Download NLTK Resources
nltk.download('punkt')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to /home/nonroot/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]   /home/nonroot/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

```
Out[2]: True
```

```
In [3]: # Stop word removal function
def remove_stop_words(text):
    # Tokenizing the text into words
    words = word_tokenize(text)

    # Defining the English stop words
    stop_words = set(stopwords.words('english'))

    # Removing stop words from the text
    filtered_words = [word for word in words if word.lower() not in
stop_words]

    return ' '.join(filtered_words)
```

```
In [5]: # Main function
if __name__ == "__main__":
    input_text = "This is an example of a text document that needs stop word
removal"
    preprocessed_text = remove_stop_words(input_text)
    print("Original Text:")
    print(input_text)
    print("\nPreprocessed Text:")
    print(preprocessed_text)
```

Original Text:

This is an example of a text document that needs stop word removal

Preprocessed Text:

example text document needs stop word removal